Chapter 5
Rethinking AI Exposure: From Occupations to Sectors

—Laura Nurski (Centre for European Policy Studies and KU Leuven) and Pierre-Alexandre Balland (Centre for European Policy Studies and Harvard Growth Lab)

Prof. Dr. Laura Nurski is Head of Programme on the Future of Work at the Centre for European Policy Studies (CEPS) in Brussels, where she leads policy research on the impact of artificial intelligence on labor markets, work organization, and job quality. She is also Assistant Professor in the Work and Organisations Studies department of the Faculty of Economics and Business at KU Leuven, where she leads the Acerta Chair AI at Work, conducting experimental (workplace) research on AI and job design. She regularly works with and for EU institutions such as the European Commission, its Joint Research Centre, and Eurofound—as well as national and regional authorities.

Prof. Dr. Pierre-Alexandre Balland is Chief Data Scientist at CEPS and Visiting Professor at Harvard Kennedy School's Growth Lab. His research focuses on economic complexity, innovation systems, and the geography of technological change, with extensive experience in linking patent, publication, and investment data to sectoral and regional economic outcomes. He has been named among Clarivate

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

Highly Cited Researchers, and his work has been influential in building data-driven industrial policy in Europe. He is also cofounder of AI World, a global observatory mapping emerging AI technologies, sectors, jobs, and innovation hubs.

Occupational lenses to AI exposure are indispensable for seeing which *jobs* contain AI-susceptible tasks—but they systematically miss where AI reshapes *workflows* and creates new tasks. A sector lens based on patents, publications, and investment surfaces where AI capabilities are being built and deployed *before* they appear in job descriptions. A comparison of both lenses shows that, while occupational measures identify AI pressure in cognitive, white-collar work, they underestimate future impact in education, health, and administrative services.

If policymakers and firms rely on occupation-level exposure scores alone, they risk allocating training funds to the wrong places (missing sectors where work is about to be reorganized) and focusing regulatory capacity on the wrong domains (underpreparing sector regulators where AI capabilities are accelerating). A readable picture of AI exposure therefore requires two signals: where *today's* occupational tasks overlap with AI capabilities, and where sectors are rapidly building and adopting AI in ways that will restructure work *tomorrow*.

Introduction

How much of your job could machines do? A decade of labor economics has converged on a way to answer that question: look at the job tasks (Acemoglu and Restrepo 2019; Autor 2013). Take detailed descriptions of what workers do, compare them to what AI systems can do, and produce an exposure score for each occupation. The resulting occupational exposure scores are now widely cited in policy and corporate planning, and their headline finding is consistent: AI exposure concentrates in white-collar, cognitive work, inverting the pattern from earlier automation waves.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

But occupational scores have a structural blind spot. Because they start from what workers currently do, they struggle to detect AI arriving "from outside" the task bundle—reorganizing workflows, creating new tasks that do not yet belong to any occupation, or shifting tasks across roles. By the time those tasks appear in occupational databases, the transformation is already complete.

The blind spot grows as AI develops into a general-purpose technology that reorganizes how work is coordinated within and across firms (Brynjolfsson et al. 2019). The question is no longer only which tasks AI can perform, but what organizational transformations it makes possible—and that is a question occupational exposure scores, by construction, cannot answer.

In this chapter, we examine what those scores reliably show and where they run out. We compare two lenses on AI exposure: an *occupational* lens, which scores how exposed each job is based on the tasks it contains, and a *sector* lens, which tracks where novel AI activity is concentrating using early signals of AI development and deployment—patents, scientific publications, and venture capital investment—that show up at the industry level before they reach occupational task descriptions.

Two Lenses on AI Exposure

Before turning to how either measure is built, this section sets out what each lens (occupational and sectoral) can see, where the occupational lens breaks down, why sectors offer a better unit of analysis, and what happens when the two are compared head-to-head.

What Occupational Exposure Can and Cannot Tell Us

Occupational exposure to AI is not a forecast of employment change, wages, or productivity. It does not tell us whether AI will be adopted in a given occupation, how quickly adoption will occur, or whether adoption will be socially or economically beneficial. What it does measure is technical feasibility: the extent to which the tasks that make up an occupation could, in principle, be affected by AI

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

systems given their current and near-term capabilities. Because such scores can be calculated for the entire labor market, they are useful for comparative analysis across occupations, sectors, regions, or demographic groups.

Exposure metrics also abstract from organizational choices, regulation, worker agency, and social norms. Two occupations with similar exposure scores may experience very different outcomes depending on how AI is implemented and whether workers shape its use. Occupational exposure is therefore a starting point for analysis rather than an endpoint—useful for identifying where attention is needed, but starting from current task content has structural limits.

Three Structural Limits of the Occupational Lens

The occupational lens has three structural limits. The first limit is that *occupational task descriptions only contain tasks that are already stabilized and institutionally recognized as part of an occupation*. Yet many of the most consequential effects of AI arise precisely through the emergence of new tasks—data labeling, model training and validation, AI governance, supervising AI output, process orchestration—that did not previously exist. These tasks do not initially "belong" to occupations; they arise within new or modified processes and value chains and are only later allocated to specific roles.

The second limit is that *occupational exposure metrics assume that task allocations to occupations stay relatively fixed*. In reality, AI adoption changes the expertise requirements of tasks, triggering their reallocation across roles—up and down the hierarchy (between associates and professionals), across seniority levels, or sideways between frontline workers and specialized support. Consider education: an occupational exposure score for teachers will capture the overlap between AI capabilities and tasks like lesson planning, assessment, or administrative tasks. What it cannot capture is a more structural reorganization in which AI systems handle individualized instruction and routine assessment while teachers move to interpreting learning analytics, making judgment calls

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

about when to override the system, and maintaining the human relationship necessary for learning.

The third limit is that *an occupational lens largely treats tasks as independent units*. Most exposure metrics implicitly assume that tasks can be automated or augmented one by one. Yet in real organizations, tasks are embedded in end-to-end processes and tightly interdependent. Whether AI substitutes or complements human work often depends less on the automatability of individual tasks than on how those tasks connect to upstream and downstream activities. This interdependence is precisely what creates the wedge between automation and augmentation, but it is largely invisible when tasks are analyzed in isolation.

Why Sectors Are a Better Unit

Four reasons argue for shifting the unit of analysis from occupations to sectors. The first reason is that *AI adoption is not uniform within occupations across industries*. A data scientist working in agriculture does something categorically different from a data scientist working in banking, even if official occupation codes treat them identically. The industry context shapes the problems to be solved, the data infrastructure available, the regulatory environment, and the competitive incentives to automate.

Second, sectors are the natural unit of production in firm-level and macroeconomic data. GDP decomposition, productivity measurement, trade flows, and investment statistics are organized primarily around industry classifications, making sectoral findings directly comparable with the mainstream empirical toolkit of applied economics.

Third, the early signals of AI activity—patents, publications, venture capital—are typically generated, funded, and deployed at the firm or industry level rather than the occupation level. A patent filed by a logistics firm to automate route planning sits naturally in Transportation and Storage. Starting from sectoral activity and asking which sectors are being transformed by AI-related IP or startup activity is both more direct and more conceptually coherent.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

Fourth, industrial policy, sectoral training programs, regulatory frameworks for AI deployment, and collective bargaining tend to operate at the sector level. Knowing that AI exposure is concentrated in public administration or financial services has direct implications for where policy attention needs to be directed.

A concrete example illustrates the logic. A bank's AI strategy doesn't live in any single occupation: data engineers build the model platform, compliance officers write oversight rules, customer-service teams get assistance from AI chatbots, and product managers redesign the underwriting workflow end-to-end. An occupational lens picks up each of these roles separately, each with different exposure scores, and misses the fact that they are all part of one coordinated sectoral transformation. The sectoral lens sees the transformation while the occupational lens sees the components.

Comparing the Two Lenses Empirically

The occupational and sectoral approaches each rank where AI matters most, but from different inputs: task descriptions inside existing occupations versus R&D activity directed at sectoral innovation. How far do these two signals agree—and what does it mean when they do not?

Figure 5.1 compares AI exposure rankings across 18 sectors: (1) occupational approaches transposed to sectors via sector-occupation employment tables, and (2) the sectoral approach, with the integrated ALL measure combining patents (PAT), publications (PUB), and startup investment (INV). The mechanics of each are detailed in the two sections that follow.

The most visible difference between the two panels is the concentration of exposure. The sectoral measure produces sharper rankings because it captures AI R&D activity directly at the sector level; the occupational approaches arrive at sector rankings indirectly—by scoring individual occupations and pivoting them to sectors—which dilutes the signal where exposure is real but unevenly distributed across the current workforce.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.



Figure 5.1 Comparing occupational and sectoral approaches

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

At the extremes, the two approaches broadly agree. Primary and secondary sectors—agriculture, mining, construction, manufacturing—score low across all measures, while knowledge-intensive services rank higher. Information and communication sit near the top on both panels. The starkest divergence appears in education, and to a smaller extent in health and administrative support, which all score higher on the sectoral approach than on the occupational ones.

Education is the clearest case. On occupational measures, teachers score moderately because much of their working time involves interpersonal tasks—classroom management, mentoring, pastoral care—that resist automation. Yet education ranks second out of 19 sectors on AI R&D intensity. Intelligent tutoring systems, adaptive assessment tools, and early warning systems for at-risk students represent an active research and investment frontier aimed not at automating what teachers currently do but at changing what teachers are for. The occupational lens captures exposure within existing jobs; it cannot see disruption that arrives from outside the current task bundle.

One occupational measure does partially bridge this gap. Loaiza and Rigobon (2024)'s augmentation score ranks education and health considerably higher than the other occupational measures. Theirs is the only one built around what humans uniquely contribute rather than what AI can replicate. That it converges with the sectoral signal is telling: the AI innovation activity concentrated in education appears to be augmenting teachers' work rather than automating their existing tasks. The implication is that how an occupational measure is constructed—whether it asks "what can AI do?" or "what do humans do that AI cannot?"—shapes whether it can see the same disruption the sectoral data picks up.

Occupational Exposure to AI

Most of what we know about AI's potential impact on the labor market comes from occupation-level exposure studies. This section walks through how occupational exposure scores are constructed

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

from task descriptions and technological capabilities, and which occupations come out as most exposed.

Data and Method

At its core, occupational exposure to AI is inferred rather than directly observed. AI adoption within firms, and especially within jobs, is uneven, poorly measured, and evolves rapidly. Exposure metrics therefore rely on linking what AI systems *can* do in theory to what workers *actually* do in their jobs. In a recent comparison of twelve studies (Nurski et al. 2025), we identify three main building blocks used to calculate occupational exposure to technology:

- Describing the task content of occupations
- Assessing the capabilities of current technologies
- Aggregating task-level exposure to the occupational level

First, we need detailed descriptions of the task content of occupations. Most studies rely on occupational databases that describe what workers do, how often they do it, and how important different tasks are within an occupation. The dominant data source is O*NET, developed by the U.S. Department of Labor and underpinning the vast majority of existing exposure measures. While O*NET offers exceptionally rich and granular job descriptions, its U.S. focus is a limitation for this field of research. Occupational task content may differ across countries because of institutional settings or sectoral composition. Even within countries, occupations can vary tremendously across firms, sectors, or workers. Using worker-level surveys such as the European Working Conditions Survey (EWCS) or the American Working Conditions Survey (AWCS) is a promising alternative to fixed job descriptions.

Second, AI's technological capabilities are inferred using different proxies for what AI and related technologies can do. Broadly, studies rely either on patents, benchmarks, or human and AI-based annotators. *Patent*-based approaches use natural language processing techniques to measure the semantic similarity between patent texts and occupational task descriptions. *Benchmark*-based

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

approaches instead draw on observed algorithm performance on well-defined tasks and datasets—such as visual question answering or reading comprehension—which are then mapped to work tasks by experts or crowdsourced workers. A third group of studies relies on *annotators* (domain experts, crowd workers, large language models, or combinations of these) to directly assess whether specific tasks can be automated or augmented. Each approach involves strong assumptions: patents may overrepresent upstream innovation rather than workplace-deployable tools; benchmarks capture narrow task performance in artificial settings; and annotation-based approaches depend on who labels tasks (often technology experts, but not occupational experts) and how questions are framed.

Third, in the final step, task-level exposure measures are aggregated to the occupational level. In most studies, occupations are treated as weighted averages of their constituent tasks, with weights based on task importance and/or frequency. Tasks deemed more susceptible to automation or augmentation contribute more strongly to the final score. Some recent approaches move beyond simple averages by incorporating dispersion in task exposure or modeling interdependencies between tasks, but these remain largely exceptions. The result is typically a continuous exposure score indicating the extent to which an occupation's task content overlaps with current or foreseeable AI capabilities.

What the Data Says

One of the most robust findings across the occupational exposure literature and the different methodologies described earlier is that AI exposure is highest in occupations with a strong cognitive and information-processing component. Contrary to earlier technologies that tended to automate routine manual work, AI exposure today is concentrated in white-collar occupations (see Figure 5.2).

The pattern across studies is consistent. Clerical and administrative workers—whose jobs center on information processing, record keeping, summarization, and standardized communication—rank among the most exposed. So do managers and professionals in

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

business, finance, law, and administration, where much of the work involves analyzing information, drafting documents, and making structured decisions: exactly the kind of cognitive labor that large language models handle well. Information and communication technologies (ICT) professionals and data analysts are also highly exposed, as their tasks strongly overlap with current AI capabilities. AI tools for coding assistance, debugging, and data analysis accelerate and extend what these workers can do.

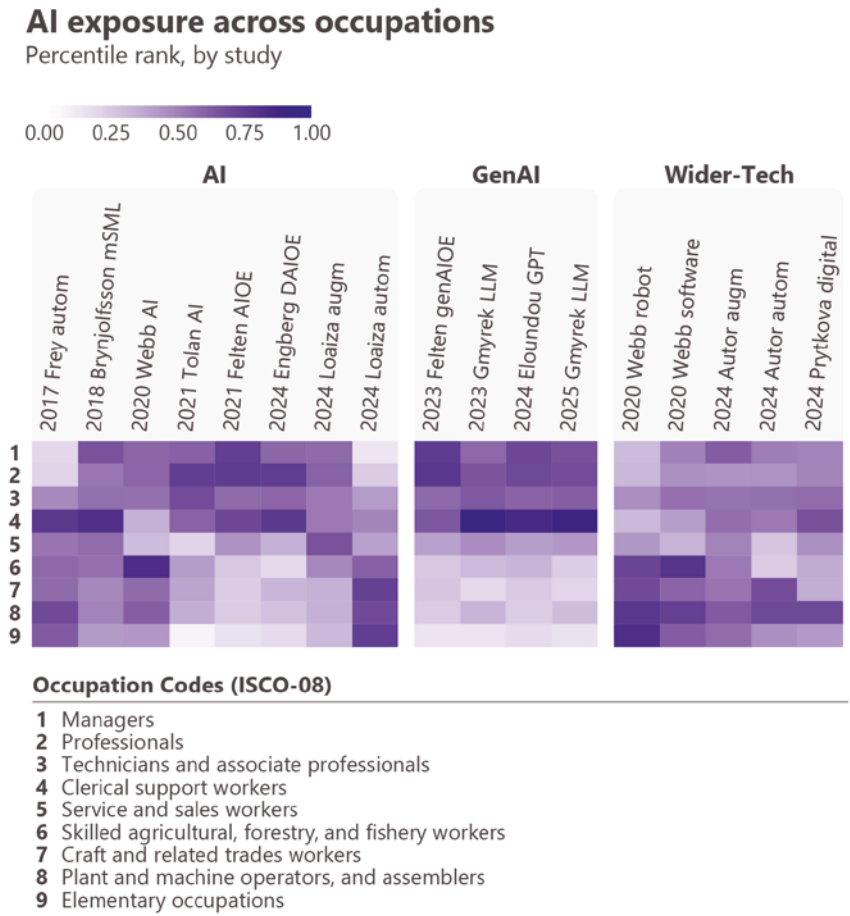


Figure 5.2 Occupational exposure scores across major occupational groups

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

By contrast, occupations with a strong physical, interpersonal, or context-specific component tend to show lower exposure. Many care-related occupations, skilled trades, and service jobs remain less exposed because their core tasks rely on physical manipulation, social interaction, or ethical judgment that is difficult to codify.

A useful way to make this concrete: a paralegal, who spends most of their day summarizing documents and drafting routine correspondence, scores very high on exposure, while a social worker conducting home visits, navigating complex family dynamics, and exercising professional judgment in unstructured environments scores low—even though both are white-collar roles.

Importantly, exposure varies substantially *within* major occupational groups. Even within professional occupations, exposure is higher for roles focused on documentation, reporting, and analysis than for roles centered on negotiation, leadership, or complex problem-solving in unstructured environments. A financial analyst building standardized models from quarterly filings scores above a portfolio manager whose job is to weigh competing strategies against each other. A radiologist reading routine chest X-rays scores above a surgeon coordinating an operating team. In each case, exposure tracks how much of the work is structured information-processing rather than judgment, coordination, or physical presence.

Demographic Gradients of Task-Based AI Exposure

Mapping AI exposure across demographic groups is one of the most common applications of occupational exposure scores, driven by policymakers' concerns about who bears the burden of AI-driven labor market change. The picture that emerges across gender, education, age, and regions is uneven—and reflects existing patterns of labor market segregation.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

Gender differences are among the most policy-relevant findings. Women are on average more exposed than men, driven by their overrepresentation in clerical, administrative, and professional service roles.

Education is strongly correlated with exposure: workers with tertiary qualifications face the highest scores.

Age differences are more modest—workers aged 25–44 show slightly higher exposure, reflecting their concentration in clerical support roles. Note that the occupational taxonomy used does not distinguish between junior and senior roles *within* occupations, which may lead to an underestimation of junior exposure.

Region correlates with exposure through the occupational mix: knowledge-intensive urban areas show higher exposure than rural or industrial regions—not because AI adoption is more advanced there, but because their occupational mix overlaps more closely with AI capabilities.

Sectoral Exposure to AI

If occupational exposure tells us which jobs contain AI-susceptible tasks, sectoral exposure asks a different question: where in the economy is AI actually being developed, funded, and deployed? This section describes the three data sources we draw on—patents, scientific publications, and venture investment—and presents the resulting ranking of 19 NACE sectors (the Statistical Classification of Economic Activities in the European Community). Together, these three early signals build an ex-ante picture of AI innovation and diffusion that complements, and in places corrects, the occupational view.

Data and Method

Measuring AI's penetration across sectors requires an ex-ante classification choice: what counts as AI, and where in the economy is it

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

showing up? We draw on three complementary sources, each capturing a different stage of the AI value chain:

- **Patents** from RegPat (Maraut et al. 2008)—a structured, internationally harmonized record of inventive activity. They are strong as an early signal, but they capture invention rather than deployment.
- **Scientific publications** from OpenAlex (Priem et al. 2022)—the flow of research output, particularly valuable for sectors where academic institutions are major knowledge producers (education, healthcare, public administration). They are skewed toward publicly funded research and slow to translate into deployed technology.
- **Venture investment** (Crunchbase Pro 2025)—forward-looking bets on commercial deployment. They are closest to near-term implementation, but they are biased toward geographies and sectors where venture capital is active.

The data cover 2020–2024 for patents and 2020–2025 for publications and venture investments. The approach (see box on Methodological Approach) combines the three streams with semantic embedding to classify documents into 19 NACE-2 sectors. Attribution is based on each document's topic and content, not the applicant's industry code: a patent whose content maps to public-administration workflows is counted there regardless of the filer's corporate identity.

Methodological Approach to Sectoral Exposure

The classification methodology is designed to maximize robustness and enrich semantic similarity with empirical validation. We operate in three main steps.

Step 1: A clean list of descriptive keywords for each sector, along with all CPC patent classes, OpenAlex scientific

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

topics, and Crunchbase industry tags, was embedded into 3,072-dimensional semantic vectors using OpenAI's text-embedding-3-large. Candidate matches were generated by cosine similarity, capturing equivalence even when terminology differs (e.g., "3D printing" vs. "additive manufacturing").

Step 2: Semantic similarity was then validated against empirical relatedness: candidate matches were retained only if CPC codes co-occurred on the same patent, topics in the same publication, or technology tags in the same funded startup. Co-occurrences were normalized using cosine similarity.

Step 3: Borderline and ambiguous cases underwent manual review to remove residual false positives, leading to a robust crosswalk between technologies and classification systems. The review is mostly used to set thresholds; results are not sensitive to hard rules.

For each sector, the derived exposure score answers a simple question: of everything the sector produces—patents, publications, or venture deals—what share is about AI? A sector scores high when AI runs through a large fraction of its activity, not when it produces many AI documents in absolute terms.

We compute AI's share of total output separately for each data source, then average the three rankings into a composite index. This is deliberately an intra-sector intensity measure. Instead of asking what fraction of all AI patents in the economy belong to a sector—which would mechanically favor large sectors—it asks how central AI is to that sector's own output, so a small AI-focused sector can score as high as a large diversified one.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

What the Data Says

The three data streams—patents, publications, and venture investment—converge on an unequal distribution of AI exposure across sectors. Combining them into a composite ranking of 19 NACE-2 sectors reveals a clear hierarchy, summarized in the bump chart presented in Figure 5.3.



Figure 5.3 AI exposure across major sector groups from patents, publications, and investments data

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

The occupational data shown in Figure 5.1 use NACE Rev. 2 (18 sectors); the sectoral data shown in Figure 5.3 use NACE Rev. 2.1 (19 sectors), which splits the former ICT sector into IT and publishing.

Rank 1: Telecommunications and IT sit at the top by every measure: patent share 36% and publication share 36.4%—both the highest of any sector—and an investment share of 27.7% that still clears the next sector by a wide margin. What registers as AI exposure elsewhere in the economy often reflects the downstream deployment of capabilities developed here. Firms build the cloud platforms, GPU clusters, and foundational models that other industries access via APIs, while internally applying AI to network fault prediction, dynamic spectrum allocation, and large-scale personalization.

Rank 2: Education ranks second in publications (25.2%) but drops in patents (21.2%) and investment (16.2%), reflecting a mismatch between research interest and commercial deployment. Public institution buyers face slow procurement cycles, tight budgets, and strict student data regulations that suppress investor appetite; and because AI in education is a highly debated academic topic, a significant share of publications addresses risks and societal implications rather than concrete deployment. The most mature applications are intelligent tutoring systems, automated assessment, early-warning systems for at-risk students, and administrative automation of admissions and scheduling—though uneven IT infrastructure and organizational conservatism constrain wider rollout.

Rank 3: Administrative and Support Services has the third-largest investment share (21%) and a notable patent share (28%) but only a modest publication share (4.57%)—a sector accumulating intellectual property (IP) around AI-augmented back-office processes without comparable research output. Use cases center on high-volume, repetitive information work: document classification, invoice processing, contract review, workforce scheduling, and customer-service routing—precisely the tasks that large language models and robotic process automation (RPA) have begun to transform at scale.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

Rank 4: Publishing, Broadcasting, and Content Production is driven by the rapid penetration of recommender systems and, more recently, generative AI into content workflows. Media companies are patenting applications for content personalization, audience analytics, and synthetic media generation; the modest publication share relative to patent intensity is consistent with a sector that innovates commercially without comparable academic output. Key applications are: text generation, image synthesis, video production automation, voice cloning for dubbing and localization, algorithmic news summarization, and AI-driven content fingerprinting for rights management.

Rank 5: Financial and Insurance Activities presents the most striking asymmetry: patent intensity at 31.6% is second only to Telecom & IT—reflecting finance's long tradition of internalizing algorithmic innovation as a proprietary advantage—yet publication share (4.10%) is among the lowest in the top half, and investment share (11.63%) is surprisingly modest. Use cases span credit scoring and anti-money-laundering monitoring in retail banking, reinforcement learning and natural language processing (NLP)-driven trading signals in capital markets, robo-advisory wealth management, and claims adjudication and parametric underwriting in insurance.

Rank 6: Public Administration & Defence combines strong patent and investment shares with near-absent publications, reflecting a mix of routine office automation and defence applications that stay outside open research channels. Civilian use cases include document processing for permits and benefit claims, fraud detection in social security, and predictive resource allocation in emergency services; defence extends to autonomous surveillance, satellite imagery analysis, real-time cyber intrusion detection, and predictive maintenance for weapons platforms.

Ranks 7–12: The middle tier shows convergent investment shares (8–14%) but divergent publication and patent profiles. Health & Social Work's low AI publication share reflects a research base dominated by clinical and biomedical topics rather than AI, even as consequential AI applications emerge in radiology, sepsis prediction,

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

AI-assisted drug discovery, and population-health risk stratification. Mining & Quarrying's relatively high publications reflect a science-heavy orientation toward geological modeling and remote sensing.
Ranks 13–19: Construction presents the sharpest internal contrast—the lowest patent share of any sector (1.14%) alongside a substantial investment share (14.73%). At the bottom, Agriculture, Accommodation & Food, and Water & Waste score low across all three sources, though Agriculture's investment share slightly outpaces its research footprint, hinting at early-stage precision-agriculture venture activity not yet matched by a comparable research base.

Conclusion

Occupational exposure scores have done useful work over the past decade. They made visible a form of technological risk that was previously hard to quantify, established that AI exposure concentrates in cognitive, white-collar work rather than manual labor, and surfaced equity-relevant patterns—women, high-skilled workers in urban areas, and early-to-mid career stages are the most exposed.

Yet occupational scores paint an incomplete picture. They cannot see AI arriving from outside the existing task bundle—not automating what workers do today, but reorganizing the work process itself. A comparison of 18 sectors shows that the two measurement approaches diverge most where this kind of structural disruption is high: in education, health, and administrative services. In those sectors, AI R&D activity is more intense than occupational task overlap would suggest. The divergence is itself the finding: AI enters these sectors through channels that escape current occupational task content.

The practical implication is straightforward: sectoral signals should sit alongside occupational exposure scores as standard inputs into workforce monitoring, training policy, and regulation. Occupational scores tell you where today's workers are most exposed. Sectoral data tells you where tomorrow's work is being built.

For workforce and training, occupational exposure points to near-term reskilling inside existing roles—tool use, verification, AI-assisted productivity—while sectoral signals identify where long-term curricula and retraining need to anticipate redesigned workflows.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.

In sectors like education and health, regulators need technical and organizational capability now, even if occupational exposure looks moderate: procurement rules, data access, and accountability standards will be stressed before workforce statistics register the change. The operating principle is the same in both: task-level upskilling for the occupations flagged by occupational scores, workflow-level capability-building for the sectors flagged by sectoral scores.

Key Takeaways

Rethinking AI exposure means holding two pictures at once—what occupational scores reveal, and what sectoral data adds. The following five takeaways trace that shift:

- Occupational exposure scores show that AI currently targets cognitive, white-collar work, inverting the pattern from earlier automation waves. They measure technical feasibility, not adoption—and should not be read as employment forecasts.
- These scores have structural blind spots: they cannot see new tasks created by AI adoption, the reallocation of tasks across organizational hierarchies, or how automating one step in a process reshapes the value of human work elsewhere in the chain.
- Sectoral data on patents, publications, and startup investment capture where AI is being developed and deployed before it shows up in job descriptions—and the sectors they flag do not always match those identified by occupational measures.
- A direct comparison across 18 sectors shows the two approaches agree at the extremes but diverge most for education, health, and administrative services—sectors where AI R&D intensity is far higher than occupational task overlap would suggest. Loaiza and Rigobon's (2024) augmentation score, built around human complementarity rather than substitution, is the only occupational measure that partially closes this gap.
- The two lenses should be used together. Workforce policy and social dialogue operate at the sectoral level, and designing them on occupational signals alone risks targeting the work of today rather than the transformation already underway.

This work is the property of John Wiley & Sons, Inc., © 2027.
This file cannot be redistributed, shared, or saved
in any way without explicit written permission from the publisher.